

Foundations and the Inclusion Exclusion Principle

The **sample space** or **universal set** has all the possible outcomes in the universe we’re working with,  $\Omega$ . An **event** is a subset of the universal set.  $A \subset \Omega$ , and an event  $x \in \Omega$  has occured iff  $x \in A$ . The **complement** of an event is the event  $A^C = \Omega \setminus A$ .

Denote  $\mathbb{A}$  as the set of all events.

Probability Measures

A Probability Measure is a map  $\mathbb{P} : \mathbb{A} \rightarrow \mathbb{R}$  such that

- $\forall A \in \mathbb{A}, \mathbb{P}(A) \geq 0$
- $\mathbb{P}(\Omega) = 1$
- for countably many disjoint events  $A_i$ , (events such that  $A_i \cap A_j = \emptyset$  when  $i \neq j$ ), it holds that  $\mathbb{P}\left(\bigcup_i A_i\right) = \sum \mathbb{P}(A_i)$

Discrete Uniform Distribution

If for all  $w \in \Omega, \mathbb{P}(\{w\}) = \frac{1}{\#\Omega}$ , then we call this a discrete uniform distribution, and  $\mathbb{P}(A) = \frac{\#A}{\#\Omega}$

Combinatorics Basics

There are  $n!$  ways to **order  $n$  objects**.  $n^k$  ways to **pull  $k$  objects with replacement**, and  $\frac{n!}{(n-k)!}$  ways to pull  $k$  objects without replacement (but in order). There are  $\binom{n}{k} = \frac{n!}{(k!)(n-k)!}$  if we ignore order (**n choose k**).

Inclusion-Exclusion Principle

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

$$\begin{aligned} \mathbb{P}(A \cup B \cup C) &= \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ &\quad - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) \\ &\quad + \mathbb{P}(A \cap B \cap C) \end{aligned}$$

Generally,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}\left(\bigcap_{k=1}^k A_{i_j}\right)$$

Conditional Probability & Bayes Theorem

Conditional Prob.: 
$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)}$$

Bayes: 
$$\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A) \frac{\mathbb{P}(A)}{\mathbb{P}(B)}$$

General Bayes: 
$$P(A_j \mid B) = \frac{\mathbb{P}(B \mid A_j) \mathbb{P}(A_j)}{\sum_{k=1}^n \mathbb{P}(B \mid A_k) \mathbb{P}(A_k)}$$

Two events are **independent**  $\Leftrightarrow \mathbb{P}(A \mid B) = \mathbb{P}(A) \mathbb{P}(B)$

Random Variables

A **random variable (RV)** is a measureable variable on the sample space  $\Omega$ . A random variable is a function  $X : \Omega \rightarrow \mathbb{R}$ , and is usually the results

of an experiment. If  $\Omega$  is every possible theoretical outcome,  $X$  lets us work with  $\Omega$  by taking real measurements.

A **cumulative distribution function (CDF)** is defined by  $F_X(x) := \mathbb{P}(X \leq x)$  and makes sense for both discrete and continuous variables, but is usually used for continuous variables.

Discrete and Continuous RVs

If there exists some index set  $I$  that is at most countably infinite such that  $\sum_{i \in I} \mathbb{P}(X = x_i) = 1$  we call  $X$  a discrete random variable. **The probability mass function** then is simply  $p(x_i) = \mathbb{P}(X = x_i)$ .

**PDF:** 
$$\mathbb{P}(X \in B) = \int_B f_{X(x)} dx$$

**CDF:** 
$$P(X \leq B) = \int_{-\infty}^x f_X(t) dt$$

Common Distributions

| Distribution               | PMF/PDF                                                                                         | $\mathbb{E}$        | Var                   |
|----------------------------|-------------------------------------------------------------------------------------------------|---------------------|-----------------------|
| Uniform on $\{1, ..., n\}$ | $\begin{aligned} \mathbb{P}(X = k) &= \\ \frac{1}{n} \text{ for } k &= 1, ..., n \end{aligned}$ | $\frac{n+1}{2}$     | $\frac{n^2-1}{12}$    |
| Bernoulli(p)               | $\begin{aligned} \mathbb{P}(X = 1) &= p \\ \mathbb{P}(X = 0) &= 1 - p \end{aligned}$            | $p$                 | $p(1 - p)$            |
| Binomial(n, p)             | $\begin{aligned} \binom{n}{k} p^{k(1-p)^{n-k}} \\ k \in \{1, ..., n\} \end{aligned}$            | $np$                | $np(1 - p)$           |
| Geometric(p)               | $\begin{aligned} \mathbb{P}(X = k) &= p(1 - p)^{k-1} \\ k \in N \end{aligned}$                  | $\frac{1}{p}$       | $\frac{1-p}{p^2}$     |
| Poisson( $\lambda$ )       | $e^{-\lambda} \frac{\lambda^k}{k!}$                                                             | $\lambda$           | $\lambda$             |
| Negative Binomial(r, p)    | $\begin{aligned} \binom{k-1}{r-1} p^r (1 - p)^{k-r} \\ k \in \{r, r + 1, ... \} \end{aligned}$  | $\frac{r}{p}$       | $\frac{r(1-p)}{p^2}$  |
| Uniform([a, b])            | $f_X = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$                                                     | $\frac{a+b}{2}$     | $\frac{(b-a)^2}{12}$  |
| Std Normal $N(0, 1)$       | $f_X = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$                                                | 0                   | 1                     |
| Gaussian $N(N, \sigma^2)$  | $f_X = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$                          | $\mu$               | $\sigma^2$            |
| Exponential( $\lambda$ )   | $f_X = \lambda e^{-\lambda x}, x \geq 0$                                                        | $\frac{1}{\lambda}$ | $\frac{1}{\lambda^2}$ |

**Uniform:** All outcomes  $1, ..., n$  are equally likely. **Bernoulli:** Success or failure, head or tail, yes or no. **Binomial:** How many successes there are out of a fixed number  $n$  of Bernoulli(p). **Geometric:** Counts number of trials till first success. **Possion:** Count of events / arrivals, e.g. calls in a

call center, patients in a hospital. **Negative Binomial:** Trials until r-th success.

**Uniform:** Continuous uniform distribution on  $[a, b]$ , i.e. every  $x$  is equally likely ( $\mathbb{1}_{[a,b]}(x)$  just checks if  $x \in [a, b]$  and returns 1 if so else 0. ) **Std.** **Normal:** Normal distribution with mean 0 and variance, e.g. standardized iid averages (**iid** = Independent and Identically Distributed.). **Gen.** **Normal:** General normal, usually assumed. **Exponential:** E.g. memoryless waiting times.

Two RVs are Independent  
$$\Leftrightarrow \mathbb{P}(X \in A \wedge Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B)$$

Expected Value

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$$

Expectations are linear:  $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$

Expectations add (even if not ind.):  $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

Variance

$$\text{Var } X = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Independence

$X$  and  $Y$  are independent  
$$\Leftrightarrow \mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$$
  
$$\Leftrightarrow \text{Var}(X + Y) = \text{Var}(X) \text{Var}(Y)$$

Linear and Monotone Transformations

If  $X, Y$  are continuous RVs,  $a, b$  are constants ( $a \neq 0$ ),  $g(x)$  is differentiable and strictly monotone function,

$$Y = aX + B \Leftrightarrow f_Y = \frac{1}{|a|} f_X \left( \frac{y - b}{a} \right)$$

$$Y = g(X) \Leftrightarrow f_Y(y) = f_X(g^{-1}(y)) \cdot \left| (g^{-1})'(y) \right|$$

Inverse function thm: 
$$(g^{-1})'(y) = \frac{1}{g'(g^{-1}(y))}$$

CDFs applied to their RVs are Uniform:  
$$Y = F_X(X) \Leftrightarrow Y \sim \text{Unif}([0, 1])$$

Central Limit Theorem

Let  $X_1, ..., X_n$  be iid random variables with  $\mathbb{E}[X_1] = \mu \in \mathbb{R}$  and  $\text{Var}(X_1) = \sigma^2 > 0$ . We define  $\overline{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ . Then,

$$\mathbb{P}\left(\frac{\overline{X_n} - \mathbb{E}[\overline{X_n}]}{\sqrt{\text{Var}(\overline{X_n})}}\right) \xrightarrow{n \rightarrow \infty} \Phi(x)$$

The CLT effectively states that for large N,

$$\frac{\overline{X_n} - \mathbb{E}[\overline{X_n}]}{\sqrt{\text{Var}(\overline{X_n})}} \Big) \overset{\text{approx}}{\sim} N(0, 1)$$

Meaning we can estimate the sampling distribution from measurements (RVs),  $X_1, ..., X_n$  using a standard normal curve!

### Joint Probability Densities

If  $X, Y$  are independent, then  $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ .

$$F_{X,Y} = \int_A \int_B f_{X,Y}(x, y) dy dx$$

### Covariance & Correlation

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

$X$  and  $Y$  are ind.  $\implies \text{Cov}(X, Y) = 0$ (conv. not always true)

$$\text{Cov}(X, Y) = \text{Cov}(Y, X) \qquad \text{Cov}(X, X) = \text{Var}(X)$$

$$\text{Cov}(aX + bY, Z) = a \text{ Cov}(X, Z) + b \text{ Cov}(Y, Z) \forall a, b \in \mathbb{R}$$

$$\text{Correlation: Cor}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

$$-1 \leq \text{Cor}(X, Y) \leq 1$$

### Cute Lemmas

$$\text{Prop 7.2.1 } \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{ Cov}(X, Y)$$

$$\text{Thm 7.22 } \text{Cov}(X, Y)^2 \leq \text{Var}(X) \text{Var}(Y)$$

### Max of Ind. RVs

Let  $X_i, i \in \{1, ..., n\}$  be iid RVs with cdf  $F_X$ . Then,

$$F_{\max_i X_i}(x) = F_X(x)^n$$

### Min of Ind. RVs

Let  $X_i, i \in \{1, ..., n\}$  be iid RVs with cdf  $F_X$ . Then,

$$F_{\min_i X_i}(x) = 1 - (1 - F_X(x))^n$$

### Sum of RVs

$$p_{X+Y}(k) = \sum_y p_X(k - y)p_Y(y)$$

$$f_{X+Y}(z) = \int_{\mathbb{R}} f_X(z - y)f_Y(y)dy$$

### Sum of Normal RVs

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

### Sums of Distributions

$$X \sim \text{Pois}(\lambda_X), Y \sim \text{Pois}(\lambda_Y) \implies X + Y \sim \text{Pois}(\lambda_X + \lambda_Y)$$

$$X \sim \text{Binom}(n_X, p), Y \sim \text{Binom}(n_Y, p) \implies X + Y \sim \text{Binom}(n_X + n_Y, p)$$

$$X \sim \text{Exp}(\lambda), Y \sim \text{Exp}(\lambda) \implies X + Y \sim \text{Gamma}(2, \lambda)$$

$$X \sim \text{Gamma}(\alpha, \lambda), Y \sim \text{Gamma}(\beta, \lambda) \implies X + Y \sim \text{Gamma}(\alpha + \beta, \lambda)$$

$$X \sim \text{Exp}(\lambda_X), Y \sim \text{Exp}(\lambda_Y) \implies \min\{X, Y\} \sim \text{Exp}(\lambda_X + \lambda_Y)$$

### Conditional Expectation

Let  $X$  be a discrete RV and A an event with  $\mathbb{P}(A) > 0$ , then

$$\mathbb{E}[X \mid A] = \sum_k x_k \mathbb{P}(X = x_k \mid A)$$

$$\text{Total Prob. for Exp. } \mathbb{E}[X] = \mathbb{E}[X|A_1]\mathbb{P}(A_1) + ... + \mathbb{E}[X|A_n]\mathbb{P}(A_n)$$

Prop 9.3  $A = n$  is divisible by k,  $B = n$  is divisible by l.

$$A \text{ and } B \text{ are independent} \iff \text{gcd}(l, k) = 1$$

### Chebyshev's Inequality

Let  $X$  be an RV with  $E[X] = \mu$  and finite variance.

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

### Weak Law of Large Numbers

Let  $(X_n)_{n \in \mathbb{N}}$  be a sequence of iid with finite variance. For any  $a > 0$ ,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq a\right) = 0$$

### Moment Generating Functions

Let  $X$  be an RV. Then, for  $n \in \mathbb{N}$  we call  $\mathbb{E}[X^n]$  the **nth-moment** of  $X$ . If  $\mathbb{E}[X]$  is finite, then we further call  $\mathbb{E}[(X - \mathbb{E}[X])^n]$  the **centered nth-moment**. This means the Expectation is the first moment and the Variance is the second centered moment.

The MGF of X, denoted by  $M_X$  is

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\sum_{n=0}^{\infty} \frac{(tX)^n}{n!}\right]$$

$$\text{Cont. Form } M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$$

$$\text{Disc. Form } M_X(t) = \sum_x e^{tx} p_X(x)$$

**To find the nth-moment**, differentiate the MGF n-times, and evaluate at 0.

If  $X_1, X_2, ..., X_n$  are independent RVs, then

$$M_{X_1 + ... + X_n}(t) = M_{X_1}(t) \cdot M_{X_2}(t) \cdot ... \cdot M_{X_n}(t)$$

| Distribution                                 | MGF                                               |
|----------------------------------------------|---------------------------------------------------|
| <b>Uniform on <math>\{1, ..., n\}</math></b> | $\frac{e^t(e^{nt}-1)}{n(e^t-1)}$                  |
| <b>Bernoulli(p)</b>                          | $1 - p + pe^t$                                    |
| <b>Binomial(n, p)</b>                        | $(1 - p + pe^t)^n$                                |
| <b>Geometric(p)</b>                          | $\frac{pe^t}{1-(1-p)e^t}$                         |
| <b>Poisson(λ)</b>                            | $\exp(\lambda(e^t - 1))$                          |
| <b>Negative Binomial(r, p)</b>               | $\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$          |
| <b>Uniform([a, b])</b>                       | $\frac{e^{tb}-e^{ta}}{t(b-a)}$                    |
| <b>Std Normal <math>N(0, 1)</math></b>       | $\exp\left(\frac{t^2}{2}\right)$                  |
| <b>Gaussian <math>N(N, \sigma^2)</math></b>  | $\exp\left(\mu t + \sigma^2 \frac{t^2}{2}\right)$ |
| <b>Exponential(λ)</b>                        | $\frac{\lambda}{\lambda-t}$                       |

### Brownian Motion

A **random walk** is a stochastic process  $\{S_n : n \in \mathbb{N}_0\}$  defined by  $S_0 = 0$  and  $S_n = \sum_{i=1}^n X_i$ , where  $(X_i)_{i \geq 1}$  are RVs.

A **brownian motion** is a stochastic process,  $\{B(t)\}_{t \in [0, \infty]}$ , satisfying:

- $W(0) = 0$  (starts at 0)
- $W(t)$  has independent increments
- For  $0 \leq s < t$ , the increment  $W(t) - W(s)$  is  $N(0, t - s)$  distributed.
- $W(t)$  has continous paths (with probability 1).

### Properties

- BM has almost surely continuous paths, but these paths are non-differentiable everywhere.
- BM is self-similar: for any constant  $c > 0$ ,  $(W(ct))_{t \geq 0}$  has the same distribution as  $(\sqrt{c} \cdot W(t))_{t \geq 0}$ .
- BM is a markov process, so the future depends only on the present.